

ACE INSTITUTE OF MANAGEMENT



**Pokhara University
Faculty of Management**



Bachelor in Computer System and Information Technology (BCSIT)

PROPOSAL ON Project “CareConnect”

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Abstract

CareConnect is a web-based Smart Donation and Volunteer Management System developed to connect donors, volunteers, and non-governmental organizations (NGOs), through a centralized digital platform. The system addresses challenges associated with traditional donation and volunteer management processes, including poor communication, lack of transparency, inefficient record management, and difficulties in volunteer coordination. CareConnect provides features such as donation management, volunteer registration, event participation, organization dashboards, reporting, and notification services.

The system is developed using modern web technologies and follows the Agile Software Development methodology. It aims to simplify social service activities by providing a secure, transparent, and user-friendly platform that enhances collaboration between stakeholders. The project is expected to improve community engagement, increase accountability, and support charitable organizations in managing their activities more efficiently.

Acknowledgement

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Abbreviations

The Abbreviations presents the shortened forms and acronyms used throughout this report along with their full meanings. It serves as a reference guide for readers, helping them understand technical terms, technologies, and organizational terminologies used in the CareConnect project. It improves readability, reduces repetition of lengthy terms, and ensures consistency in the documentation by providing a clear explanation of commonly used abbreviations.

Abbreviation	Meaning
NGO	Non-Governmental Organization
UI	User Interface
DBMS	Database Management System
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
SDLC	Software Development Life Cycle
CRUD	Create, Read, Update and Delete

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Chapter 1: Introduction

1.1 Background

CareConnect is a web-based Smart Donation and Volunteer Management System designed to connect donors, volunteers and charitable organizations through a single digital platform. In today's world, many people are interested in helping others and supporting social causes such as orphanages, old age homes, disaster relief programs, and community welfare campaigns. However, people often face difficulties in finding a trusted and organized platform where they can easily donate money, provide necessary items, or participate in volunteer activities. Similarly, many NGOs and charity organizations struggle to manage donations, coordinate volunteers, maintain records, and communicate effectively with supporters.

With the rapid growth of internet access and digital technology, online platforms have become more practical, accessible, and efficient for social service activities. People now prefer digital systems that are fast, secure, transparent, and easy to use. CareConnect aims to use technology to bridge the gap between donors, volunteers, and organizations by providing a centralized platform where all activities can be managed smoothly.

The main purpose of CareConnect is to create a secure, transparent and reliable environment for social contribution. Through this platform, users will be able to register as donors or volunteers, donate money or items, join social events and campaigns, and track their activities in real time. Organizations will also be able to post their needs, manage volunteers, monitor donations, and maintain proper records efficiently. The system will include features such as user registration, donation management, volunteer coordination, event participation, notifications, and reporting.

CareConnect not only helps organizations reduce manual work and improve efficiency, but also encourages more people to participate in social causes by making the process simple and convenient. By improving communication, transparency, and coordination among all stakeholders, the platform aims to increase social impact and build stronger community support.

1.2 Statement of Problem

Many charity organizations, NGOs, orphanages, and social service groups still use traditional and manual methods to manage donations, volunteers and events. These methods are often slow, difficult to manage, and less effective, especially when handling a large number of activities and users.

The major problems faced by existing systems are:

1. Lack of Proper Communication
 - ❖ Poor coordination between donors, volunteers, and organizations
 - ❖ Difficulty in managing announcements and notifications
2. Manual and Inefficient Record Management
 - ❖ Donation records are often managed manually
 - ❖ High chances of data errors, duplication, and loss of information
3. Lack of Transparency
 - ❖ Donors cannot easily track how their donations are used
 - ❖ Reduced trust and confidence among users
4. Volunteer Management Difficulties
 - ❖ Volunteers face problems finding suitable campaigns or events
 - ❖ Organizations find it difficult to manage volunteer schedules and participation

CareConnect aims to solve these problems by providing a Smart Donation and Volunteer Management System that connects donors, volunteers, and organizations through a single integrated web platform. The system will improve transparency, communication, coordination, and resource management while reducing manual work and increasing social impact.

1.3 Objectives of the Study

The main objective of the CareConnect Smart Donation and Volunteer Management System is to design and develop an integrated digital platform that improves the way donations, volunteers, and charitable organizations interact and coordinate. The system aims to make social service activities more efficient, transparent and user-friendly by using modern web technologies.

The specific objectives of this project are:

1. To develop a centralized platform
2. To connect all stakeholders in one system
3. To simplify the donation process
4. To support organizations in managing needs
5. To improve volunteer coordination
6. To ensure transparency and accountability
7. To enhance organization of social services
8. To improve communication and engagement

1.4 Scope of the Study

The CareConnect project focuses on creating a platform that connects donors, volunteers, and NGOs in one system. It mainly includes donation management, volunteer registration, NGO activity handling, and event participation. Users can securely register and log in, allowing different roles like donor, volunteer, and NGO to access relevant features easily.

The system also provides reporting, communication, and notification services to improve coordination and transparency. NGOs can manage activities and track progress, while users receive updates about donations, events, and announcements. Overall, the scope is to make social welfare activities more organized, efficient, and accessible through a digital platform.

1.5 Limitations of the study

- The study is limited by time constraints, which restricted deeper research and full feature development.
- It is based on a student-level project, so it does not include all advanced or enterprise-level functionalities.
- Limited resources (tools, data, and technologies) affected the overall depth and scope of the system.
- The system uses simulated or limited test data, which may not fully represent real-world usage.
- Real-time processing and advanced analytics features are either limited or not fully implemented.
- Scalability testing and handling of large user loads were not extensively performed.

Chapter 2: Literature Review

2.1 Fundamental Theories and Concepts

2.1.1 Web-Based Information Systems

CareConnect is based on a web-based information system architecture, which allows users to access services through a browser without installing additional software. Such systems typically use client-server architecture where the frontend handles user interaction and the backend manages data processing and storage.

2.1.2 User Authentication and Authorization

Authentication ensures that only registered users can access the system, while authorization controls what each user is allowed to do. This is essential in CareConnect to distinguish between donors, volunteers, and administrators.

2.1.3 Donation Management Systems

Donation management systems are platforms designed to facilitate transparent and efficient handling of donations. They help track contributions, manage donors, and ensure proper allocation of resources.

2.1.4 Volunteer Management Systems

Volunteer management systems help organizations register, schedule, and coordinate volunteers for different activities and events. These systems improve communication and resource allocation.

2.1.5 Notification and Communication Systems

Modern web applications integrate email or messaging services to send notifications. CareConnect uses such mechanisms to inform users about events, donations, and updates.

2.2 Review of Similar Projects and Systems

To understand the current approaches used in donation and volunteer management, several existing platforms were studied:

GoFundMe

<https://www.gofundme.com/>

GoFundMe is a global crowdfunding platform used mainly for personal fundraising and charitable causes. Users can create campaigns to raise money for medical emergencies, education, disaster relief, and social causes. It allows individuals to share campaigns through social media and receive donations online. However, it is mostly focused on money-based fundraising, and has limited structured volunteer or organization management features.

VolunteerMatch

<https://www.volunteermatch.org/>

VolunteerMatch is a platform designed to connect volunteers with nonprofit organizations. It allows organizations to post volunteer opportunities, and individuals can search and apply based on location, interest, and skills. However, it mainly focuses on volunteering only, and does not support donation collection or integrated fundraising tools.

Salesforce Nonprofit Cloud

[Salesforce Nonprofit Cloud | Salesforce](#)

Salesforce Nonprofit Cloud is a comprehensive management system for large nonprofit organizations. It provides tools for donor management, fundraising tracking, campaign management, and analytics. It is very powerful but is often considered expensive and complex to implement, making it more suitable for large-scale organizations rather than small NGOs or community groups.

Charity Navigator

<https://www.charitynavigator.org/>

Charity Navigator is a platform that evaluates and rates nonprofit organizations based on financial health, transparency, accountability, and performance. It helps donors decide which charities are trustworthy. However, it does not provide fundraising or volunteer coordination features; it is mainly an information and rating system for donors.

2.3 Research Gap

From the review of existing systems, several gaps have been identified:

- Lack of unified platforms combining donation, volunteer, and event management.
- Limited transparency in local NGO systems.
- Poor user experience in many existing systems.
- Lack of real-time communication and notification features.
- Minimal reporting and analytics features for NGOs

CareConnect aims to address these gaps by providing a centralized, user-friendly, and transparent platform.

Chapter 3: System Analysis and Design

3.1 Requirement Analysis

The CareConnect system is designed as a web-based donation and volunteer management platform that connects donors, volunteers, and NGOs. The main requirements were identified through analysis of existing manual and semi-digital donation systems.

3.1.1 Functional Requirements

Functional requirements describe the main services and operations that the system performs.

1. User Registration and Role Management
 - Users can register and log in securely
 - Role-based access for Donor, Volunteer, NGOs/ Organization and Admin
 - Different dashboards based on user roles
2. Donation Management
 - Users can donate money and essential items
 - Donation submission and tracking system
 - Maintain donation history and records
3. Volunteer Registration
 - Volunteers can register and create profiles
 - Apply for available volunteer opportunities
 - Track participation in events

4. Organization Dashboard

- Organizations can manage their activities
- Post donation requests and volunteer opportunities
- Track donations and volunteer applications

5. Event Management

- Create and manage events
- Users can join and participate in events
- Track event progress and status

6. Transparent Donation Records and Reports

- Maintain detailed records of all donations
- Generate reports for accountability and transparency

7. Dashboard and Analytics

- Donor dashboard to view contributions and history
- Admin dashboard with system overview and insights
- Display basic analytics such as donations and participation trends

8. Communication and Notification System

- Email notifications to donors about donations and updates
- Alerts for events and activities

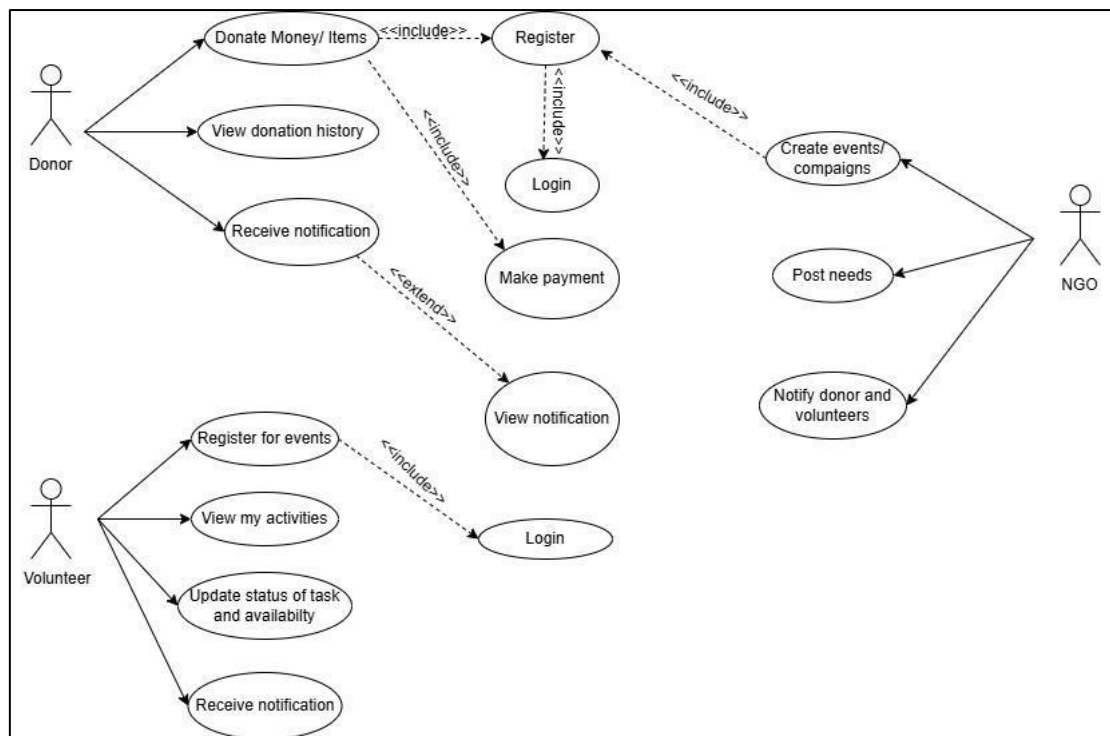


Figure 1: Use Case Diagram

3.1.2 Non-Functional Requirements

Non-functional requirements define the quality and performance aspects of the system.

1. Security
 - Secure authentication system with password protection
 - Role-based access control to prevent unauthorized access
2. Performance
 - Fast response time for user requests
 - Efficient handling of multiple users and transactions
3. Usability
 - Simple and user-friendly interface
 - Easy navigation for donors, volunteers, and organizations
4. Reliability
 - System ensures accurate donation and volunteer records
 - Minimal system downtime
5. Scalability
 - System can handle increasing number of users and data
 - Can be expanded with new features in future
6. Maintainability
 - Easy to update and modify system features
 - Modular design for better maintenance

3.2 Feasibility Analysis

1. Technical Feasibility

The system is technically feasible using modern web technologies such as HTML, CSS, JavaScript, and backend frameworks like Node.js or PHP with MySQL database support.

2. Operational Feasibility

The system is user-friendly and can be easily operated by donors, volunteers, and NGOs with minimal training.

3. Financial Feasibility

The project requires minimal cost as it uses open-source technologies and existing development tools.

4. Schedule Feasibility

The system can be developed within the academic project timeline using phased development (analysis, design, implementation, testing).

3.3 Data Modeling

The data model for CareConnect is designed to support a unified platform where NGOs, donors, and volunteers interact. It captures user management, fundraising campaigns, monetary and in-kind donations, volunteer engagement, and public inquiries. The schema ensures traceability of donations, verification of NGOs, and role-based access.

Key Entities and Relationships:

1. Users serve as the central identity, holding login credentials and role information (e.g., donor, NGO admin, platform admin).
2. NGOs are linked to users who manage them, storing registration details, focus area, verification status, and contact information.
3. Causes represent fundraising campaigns created by NGOs, tracking financial goals and amounts raised.
4. Donations record monetary contributions from users to specific causes, including payment status and frequency (one-time or recurring).
5. Item Donations handle physical goods, capturing item descriptions, meetup logistics, and status.
6. Volunteers store user availability, skills, and application status for NGO opportunities.
7. Contact Messages capture public inquiries and newsletter subscriptions.
8. A separate token table (appears as repeated index entries) manages password resets.

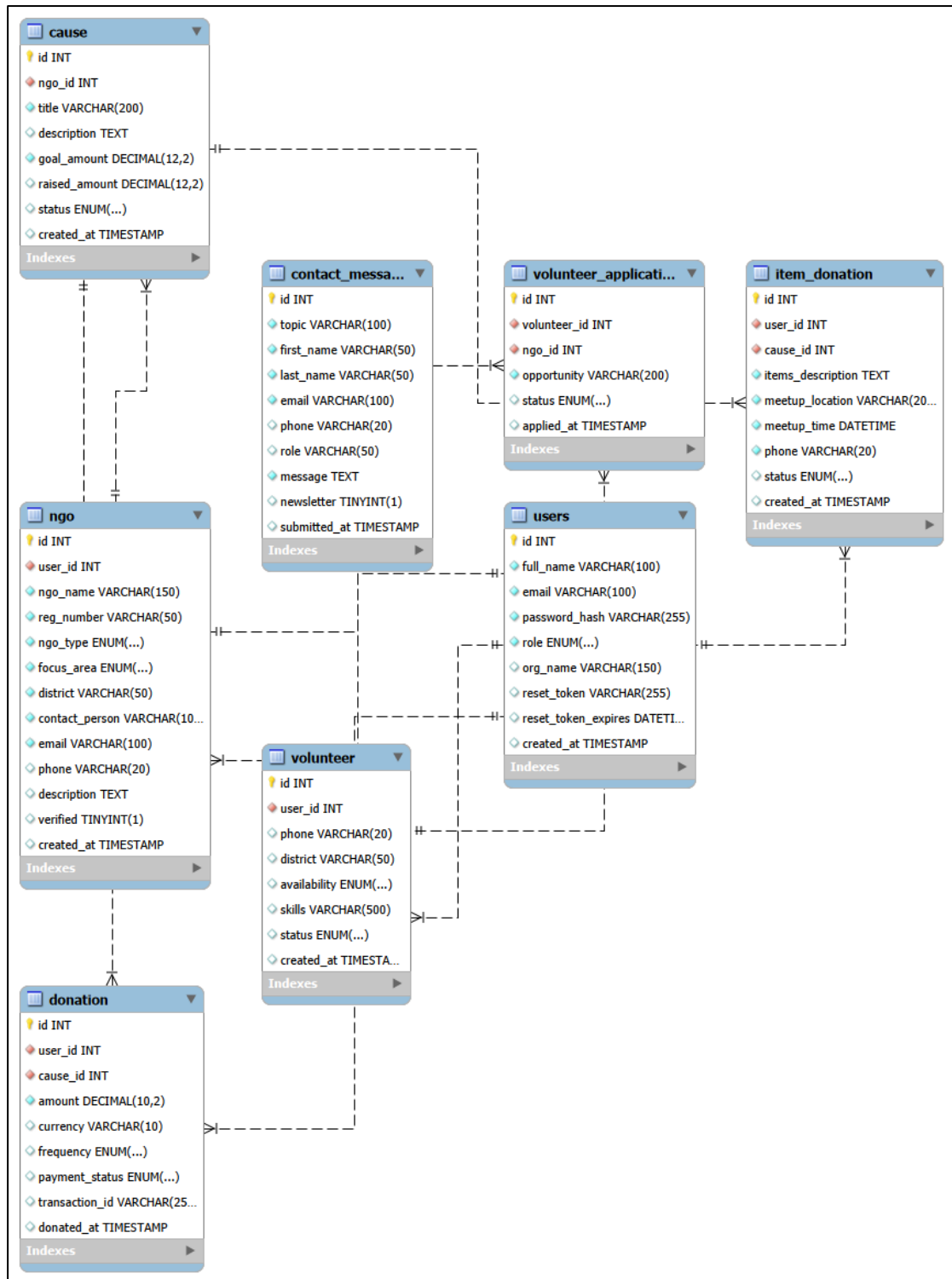


Figure 2: ER Diagram

3.4 Process Modeling

Process modeling describes system workflows:

- User registration → login → dashboard access
- Donor selects donation type → submits donation → confirmation
- NGO creates event → volunteers register → participation tracking
- Admin monitors system activity and generates reports

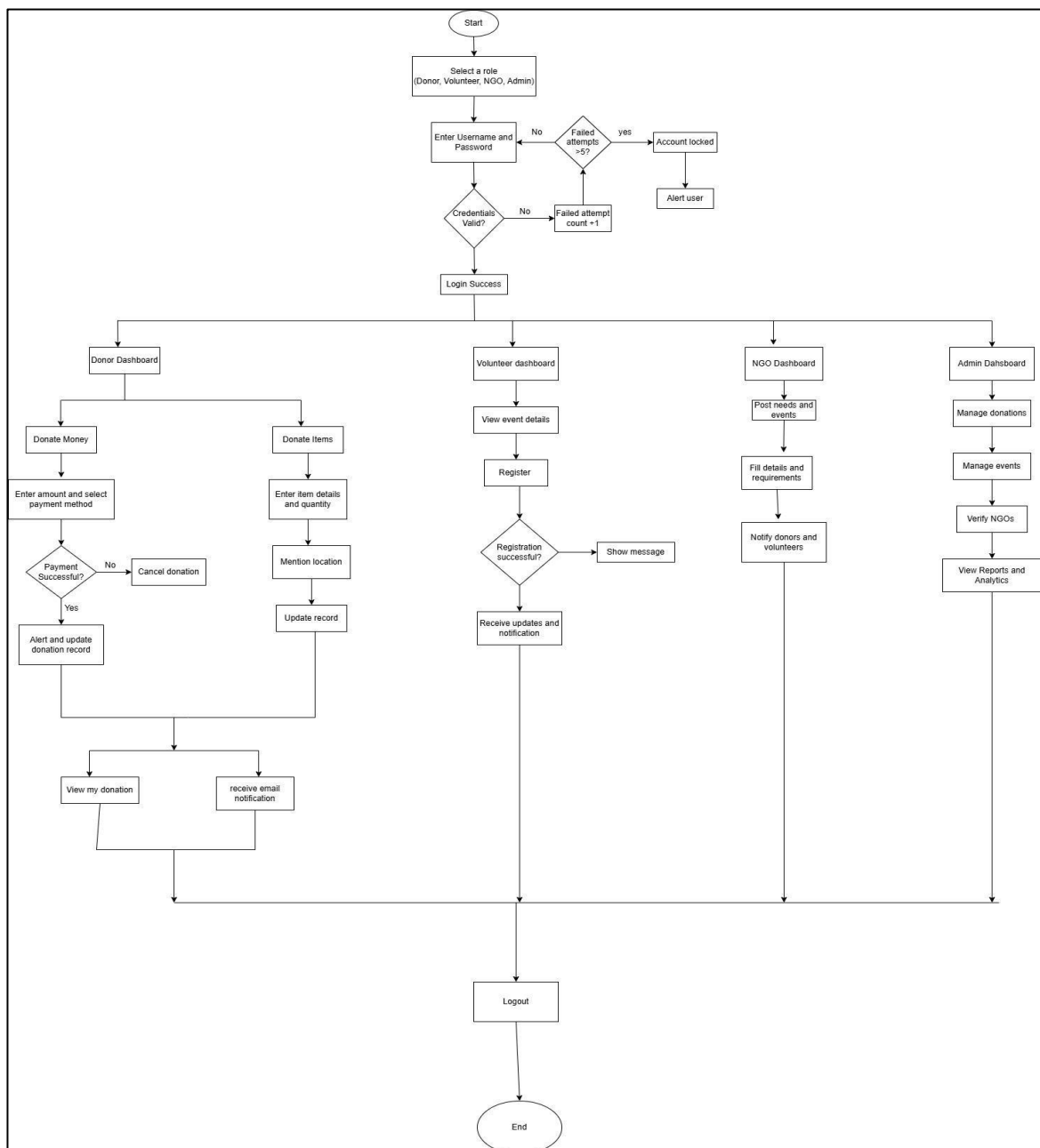


Figure 3: Flowchart

3.5 System Design

3.5.1 Architectural Design

CareConnect follows a 3-tier architecture:

- Presentation Layer (Frontend: HTML, CSS, JavaScript)
- Application Layer (Backend logic and processing)
- Data Layer (Database: MySQL)

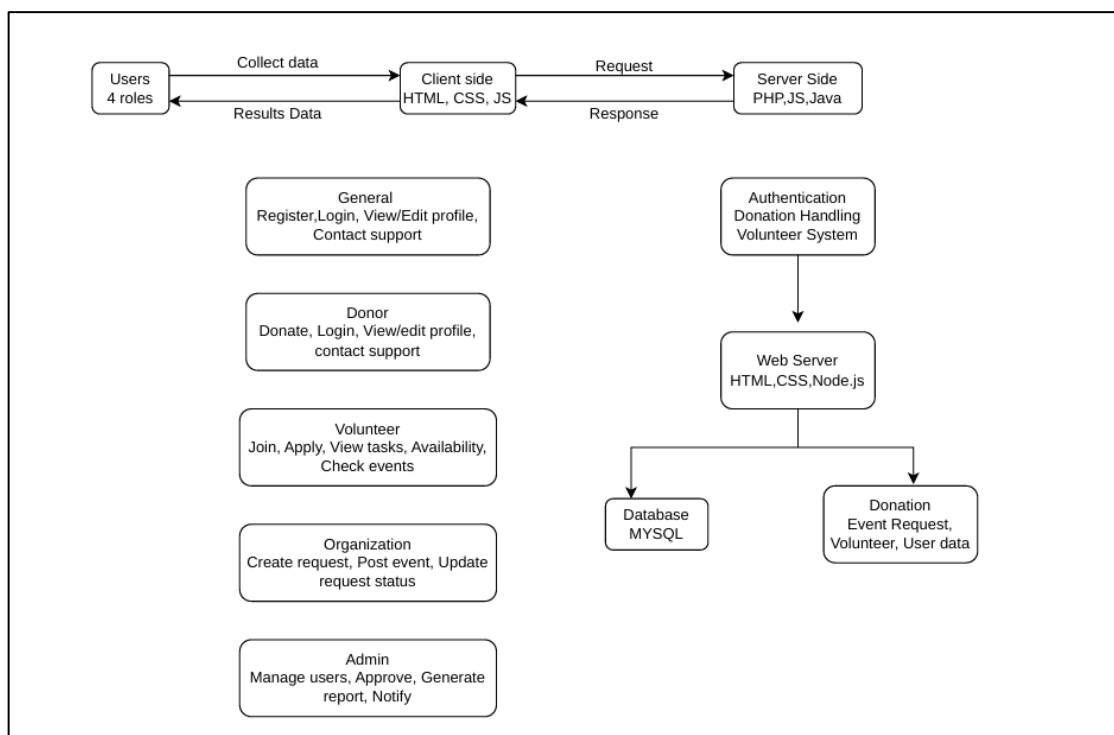


Figure 4: System Architecture

Chapter 4: Implementation and Testing

4.1 Implementation

The CareConnect project is currently in the implementation phase. At this stage, the system includes a fully designed user interface (UI) and a completed database schema. The backend logic and full system integration are still under development.

The implementation focuses on:

- Designing responsive web pages for users (donor, volunteer, NGO)
- Creating and structuring the relational database
- Preparing system modules for future backend integration

4.2 Tools Used

The following tools and technologies were used in the current phase:

- Frontend Technologies: HTML5, CSS3, JavaScript
- Database Platform: MySQL (via MySQL Workbench / phpMyAdmin)
- Editor: Visual Studio Code
- Browser: Chrome (testing UI)

4.3 Implementation Details of Modules

At this stage, only the UI and database modules are implemented.

1. User Interface Module

- Login and registration pages designed
- Dashboard layouts for donor, NGO, and volunteer
- Navigation bar and homepage structure completed

2. Database Module

- Tables created for users, NGOs, donations, causes, volunteers, and messages
- Relationships defined using foreign keys
- Data structure prepared for system integration

3. Donation UI Module

- Donation form UI created
- Input fields prepared for future backend processing

4. Volunteer UI Module

- Volunteer registration form designed
- Application UI for NGOs created

4.4 Testing

Testing has been performed only on the frontend and database structure since backend integration is not yet completed.

Table 1

Test Case	Description	Result
UI Load Test	Check if all pages load correctly	Passed
Form Input Test	Check validation of forms	Partially Passed
Navigation Test	Check page linking	Passed
Database Connection Test	Verify tables creation	Passed

Chapter 5: Conclusion and Future Recommendations

5.1 Lesson Learnt

During the development of the CareConnect project, several important concepts were learned, including:

- Designing user-friendly interfaces using HTML, CSS, and JavaScript
- Structuring relational databases using MySQL
- Understanding system design concepts such as ER diagrams and data modeling
- Importance of modular development and planning before implementation

5.2 Conclusion

The CareConnect project is currently in its early development stage. The UI design and database structure have been successfully completed. Although the system is not fully functional yet, the foundation has been properly built for future backend integration and system development.

The project aims to provide a platform for donation management, volunteer coordination, and NGO activities in a centralized system.

5.3 Recommendations

To improve and complete the system in the future, the following improvements are suggested:

- Development of backend functionality using PHP or Node.js
- Integration of secure authentication system
- Payment gateway integration for donations
- Real-time notification system
- Improved UI/UX for better user experience

Appendices

In this section, screenshots of the CareConnect project are included to provide visual evidence of the system's development and functionality. These screenshots represent different parts of the project such as the user interface, including the homepage, login and registration pages, and donation and volunteer modules.

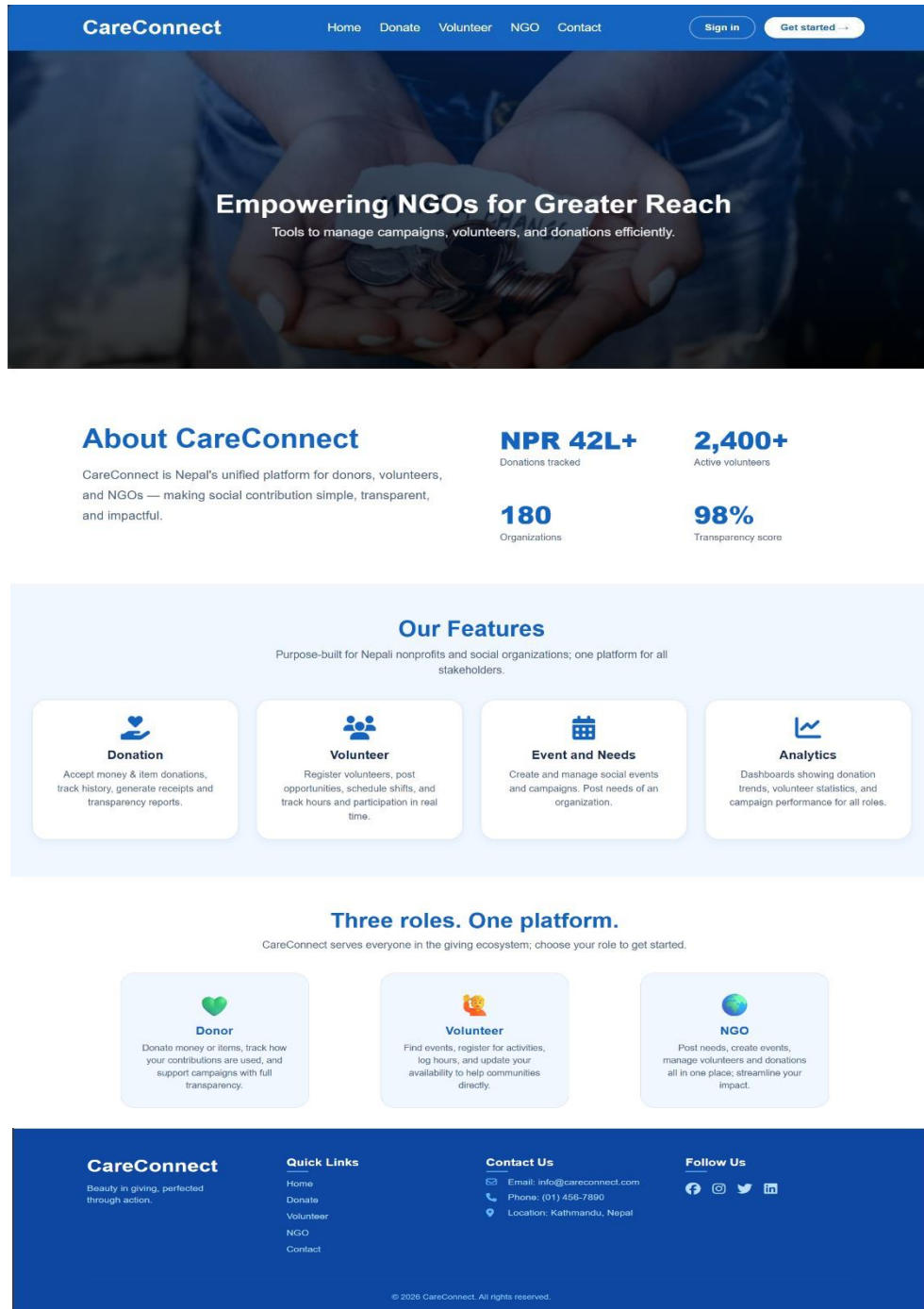


Figure 5: Home Page

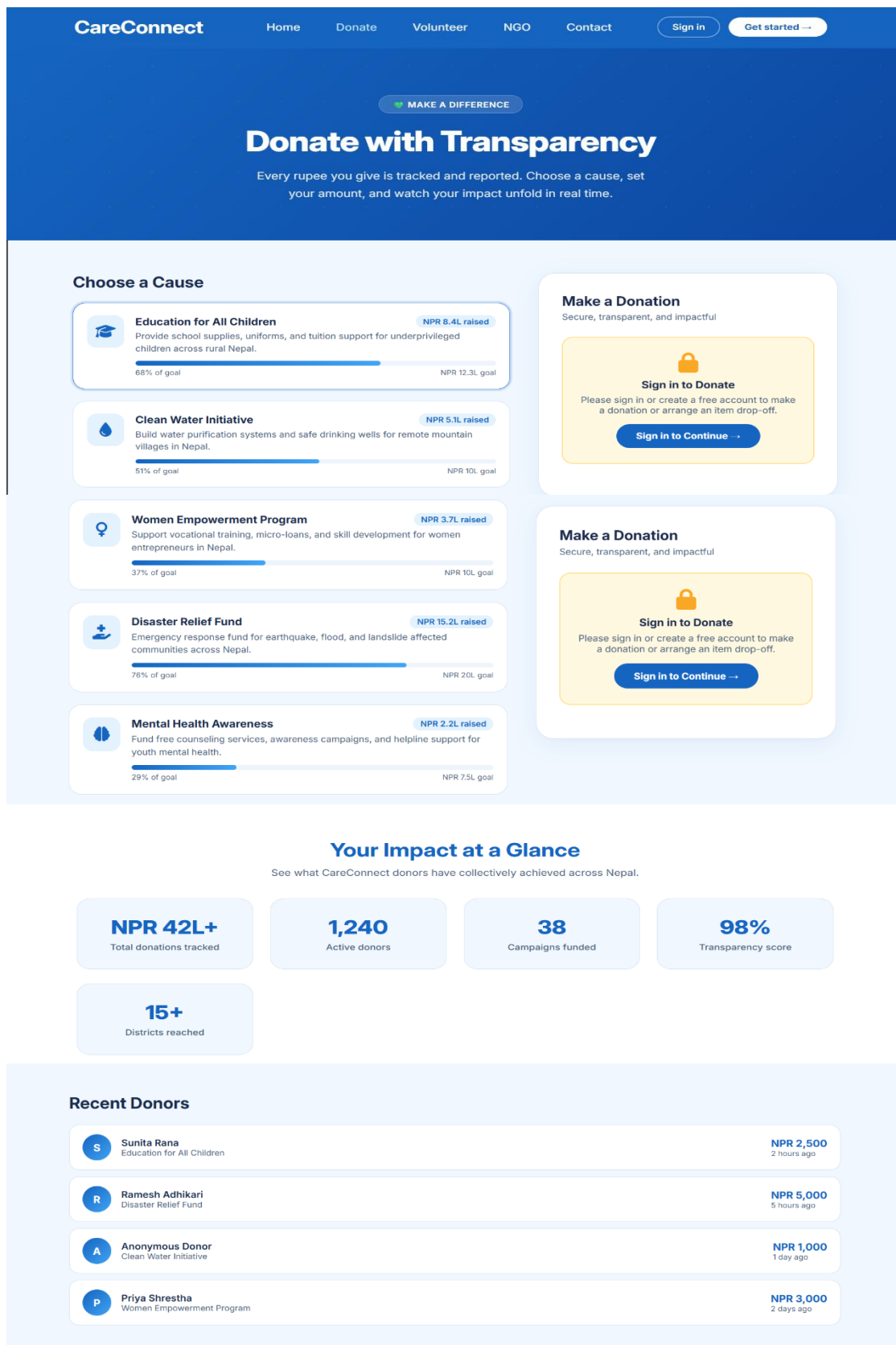


Figure 6: Donate Page

CareConnect
Home
Donate
Volunteer
NGO
Contact

Sign in
Get started →

BE A CHANGEMAKER

Volunteer Your Time, Transform Lives

Discover meaningful opportunities across Nepal. Register, apply, and track your volunteer hours — all in one place.

2,400+
Active Volunteers

180+
Opportunities Posted

45,000+
Hours Contributed

15
Districts Covered

Please sign in to apply for opportunities
You can browse opportunities, but you need to [sign in](#) or [create an account](#) to submit a volunteer application.

H
HopeNepal Foundation
NGO · Verified
Urgent

School Teaching Support — Sindhupalchok

Assist teachers in public schools with English and Math classes. Help children aged 7–14 improve their literacy and numeracy skills.

Teaching
Education
Rural

Sindhupalchok
Jun 15 – Jul 30

4 hrs/week
No experience needed

8 spots left
Apply Now

N
Nepal Health Initiative
NGO · Verified

Free Health Camp — Volunteer Medic

Join our mobile health camp team providing basic health checkups, blood pressure, diabetes screening, and medicine distribution in remote villages.

Medical
Health
Camp

Kavre District
Jun 20 – Jun 22

3 days
Medical background preferred

15 spots left
Apply Now

G
Green Nepal
NGO · Verified

Community Tree Plantation Drive

Plant 5,000 trees across urban Kathmandu parks and river banks. Help restore green cover and reduce urban heat island effects.

Environment
Outdoor
One-day

Kathmandu
Jun 5, 2026

1 day
Open to all

40 spots left
Apply Now

Become a Volunteer

Create your volunteer profile once and apply to multiple opportunities. Track your hours, earn certificates, and build your social impact portfolio.

- Verified certificates** — Get a CareConnect volunteer certificate for completed programs.
- Hour tracking** — Automatically log and track all volunteer hours in your dashboard.
- Smart matching** — Get notified about opportunities that match your skills and availability.
- Community** — Connect with 2,400+ like-minded volunteers across Nepal.

Sign in to Register as a Volunteer

You need to be signed in to create a volunteer profile and apply for opportunities. Sign in or create a free account to get started.

Sign in to Register →

Figure 7: Volunteer Page

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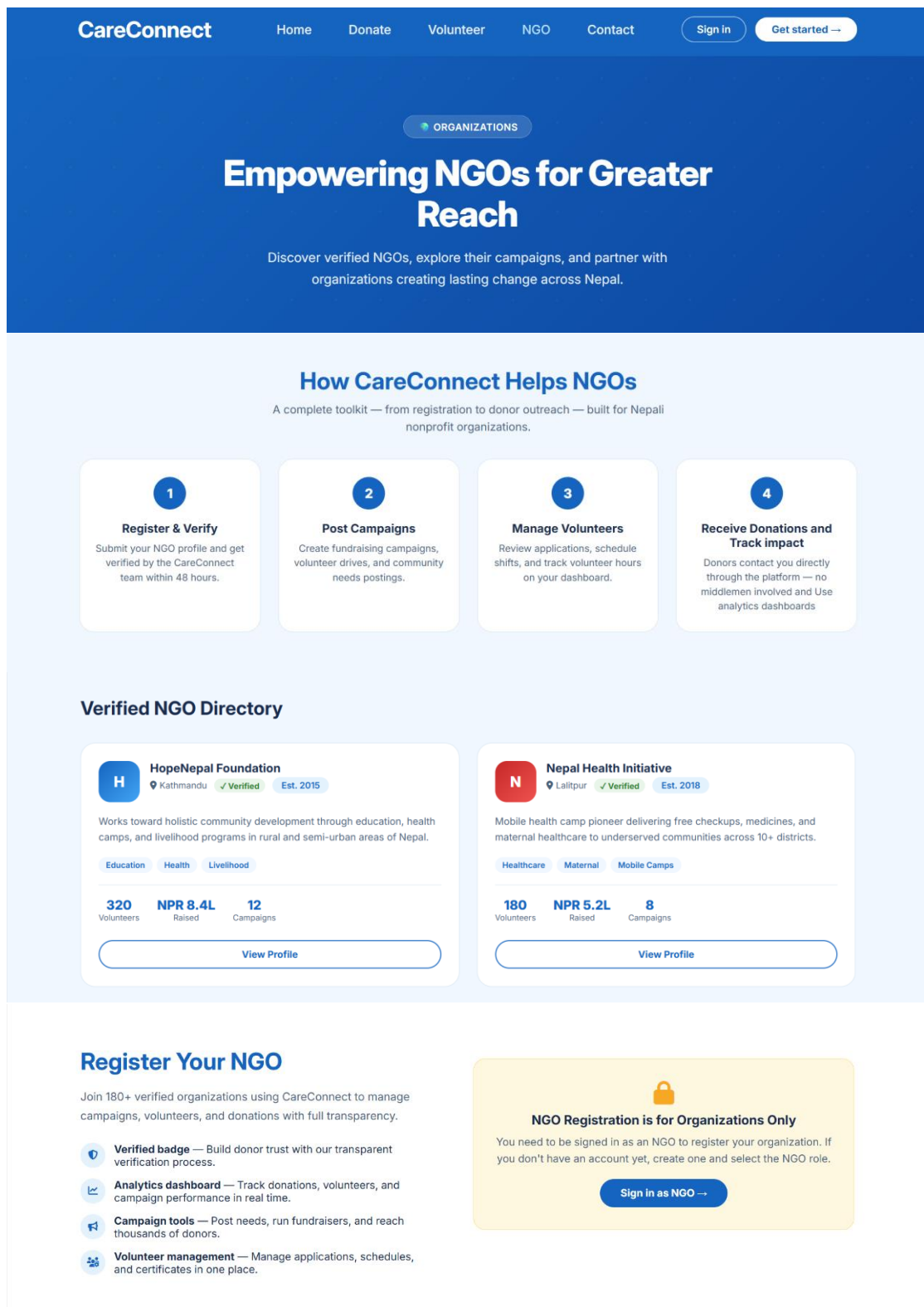


Figure 8: NGO Page

CareConnect

[Home](#)
[Donate](#)
[Volunteer](#)
[NGO](#)
[Contact](#)

[Sign in](#)
[Get started →](#)

GET IN TOUCH

We'd Love to Hear From You

Have a question, partnership idea, or feedback? Reach out to the CareConnect team — we typically respond within 24 hours.

Contact Information

We're here to help donors, volunteers, and NGOs get the most out of CareConnect. Reach us through any of the channels below.

Email Us

General: info@careconnect.com
 NGO support: ngo@careconnect.com

Call Us

Main office: +977 01-456-7890
 WhatsApp: +977 9800 000 000

Office Hours

Sunday – Friday	9:00 AM – 6:00 PM
Saturday	10:00 AM – 2:00 PM
Public Holidays	Closed

Connect on Social Media

[Facebook](#)
[Instagram](#)
[LinkedIn](#)

Send Us a Message

Fill in the form below and we'll get back to you as soon as possible.

What's this about?

Donation support

Volunteering

NGO partnership

Report a bug

Feature request

General inquiry

First name *

Last name *

John

Doe

Email address *

Phone (optional)

you@example.com

+977 98XXXXXXX

Role

I am a...

Message *

Tell us how we can help you...

0/1000 characters

☐
 Subscribe to our newsletter for updates and impact reports

Send Message

Average response time: under 24 hours

Frequently Asked Questions

Quick answers to common questions from our community.

How do I donate to a specific cause?

Is my donation secure?

How do I register as a volunteer?

How long does NGO verification take?

Can I donate physical items instead of money?

Is CareConnect free to use?

How can I track my past donations?

Can I volunteer from outside Nepal?

Figure 9: Contact Page

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CareConnect

Every act of giving matters

Nepal's smart platform for donors, volunteers, and organizations.

×

Welcome back

Sign in to your CareConnect account

Email address

Password

Sign in as

Select your role

☒ Remember me
 [Forgot password?](#)

Sign in →

Don't have an account? [Create account →](#)

Figure 11: Sign in module

CareConnect

Join the CareConnect family

Create your free account and start making a difference.

^

Create account

Join CareConnect and start your journey

Full name

Email address

Password

Minimum 8 characters

Confirm password

I want to join as

Donor
Give & track donations

Volunteer
Help & serve communities

NGO / Organization
Manage campaigns

☐ I agree to the [Terms](#) and [Privacy Policy](#)

Create my account →

Figure 10: Sign Up Module

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